

Retaining Wall Design & Verification Report

RT Wall Cantilever

Date: July 18, 2026

This engineering report compiles the retaining wall dimensions, material properties, applied loads, stability safety factors, bearing capacity analysis, FEM settlement profiles, and pile foundation design checks.

1. Design Inputs & Parameters

Wall Geometry:

Parameter	Symbol	Value
Out-of-plane thickness	t	1.000 m
Wall height	H_w	3.000 m
Toe length	L_{toe}	0.800 m
Heel length	L_{heel}	0.800 m
Top wall thickness	b_{top}	0.300 m
Bottom wall thickness	b_{bot}	0.400 m
Footing thickness	h_{ftg}	0.300 m

SNI Typical Geometry Verification Checklist:

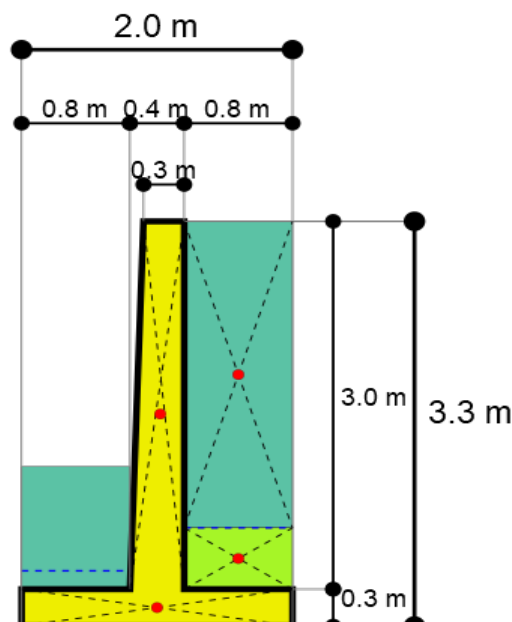
Parameter Name	Symbol	Value	SNI Requirement	Status
Top Wall Thickness	b_{top}	0.30 m	≥ 0.30 m	PASS
Base Stem Thickness	b_{bot}	0.40 m	≥ 0.33 m	PASS
Front Batter Slope	$slope$	0.0333	≥ 0.0208	PASS
Footing Width	B	2.00 m	1.32 ~ 2.31 m	PASS
Footing Thickness	h_{ftg}	0.30 m	0.27 ~ 0.33 m	PASS
Toe Slab Length	L_{toe}	0.80 m	≥ 0.67 m	PASS

Soil & Water Parameters:

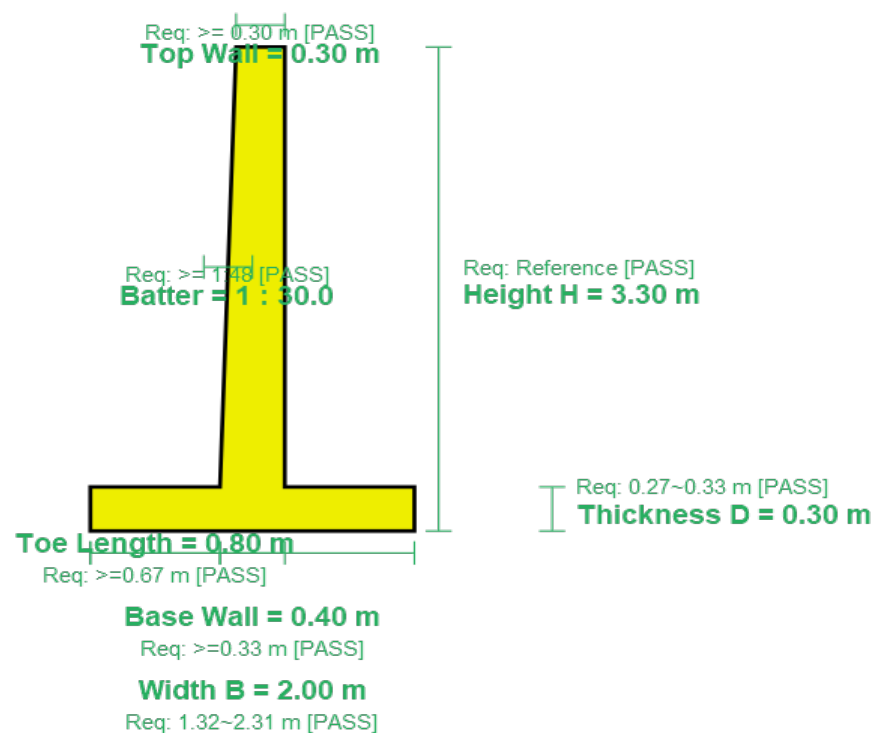
Parameter	Symbol	Value
Dry unit weight	γ_{dry}	18.00 kN/m ³
Wet unit weight	γ_{wet}	20.00 kN/m ³
Friction angle	ϕ	45.0°
Soil cohesion	c	10.0 kPa
Surcharge load	q	10.00 kPa
Soil height above heel	h_{soil}	3.000 m
Soil above toe	h_{toe}	1.000 m
Water unit weight	γ_w	9.81 kN/m ³
Water height behind wall	H_{wtr}	0.50 m
Water height in front	$H_{wtr,front}$	0.150 m

2. Model & Geometry Diagrams

Wall Profile & Mesh Grid:



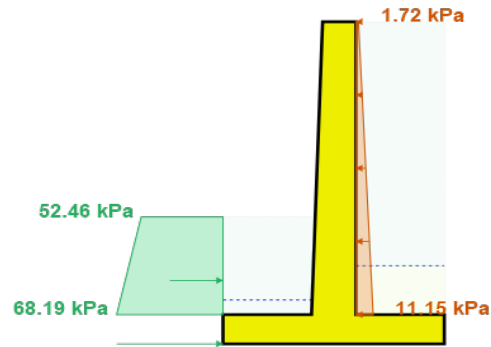
SNI Geotechnical Dimensions Proportions Verification:



3. Applied Loads & Pressure Distributions

The pressure diagrams and corresponding analytical safety check calculations are shown below:

A. Soil Lateral Pressure (Active & Passive):



- Active Earth Pressure Coefficient Equation:

$$K_a = \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$$

Result: $K_a = 0.1716$

- Active Pressure (Top of Stem):

$$p_{act, top} = K_a \cdot q$$

Result: $p_{act, top} = 1.72 \text{ kPa}$

- Active Pressure (Bottom of Footing):

$$p_{act, bot} = K_a \cdot (\gamma_{dry} z_{dry} + \gamma_{wet} z_{wet}) + K_a \cdot q$$

Result: $p_{act, bot} = 11.15 \text{ kPa}$

- Passive Earth Pressure Coefficient Equation:

$$K_p = \tan^2 \left(45^\circ + \frac{\phi}{2} \right)$$

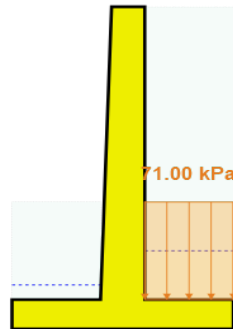
Result: $K_p = 5.8284$

- Passive Pressure (Bottom of Footing):

$$p_{pass, bot} = \frac{K_p \cdot \gamma_{dry} \cdot (h_{toe} + h_{fig})}{FS_{pass}}$$

Result: $p_{pass, bot} = 68.19 \text{ kPa}$ (Mobilized, $FS_{pass} = 2.0$)

B. Soil Vertical Weight & Surcharge:

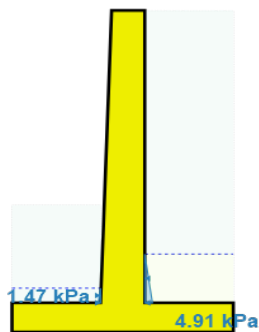


• Vertical Backfill Pressure Equation:

$$p_{v,heel} = q_{soil} + q$$

Result: $p_{v,heel} = 71.00 \text{ kPa}$

C. Hydrostatic Water Pressure:

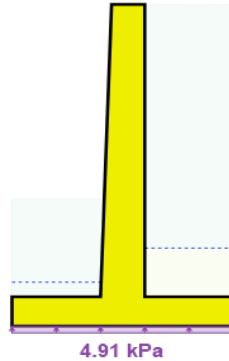


• Hydrostatic Water Pressure Equation:

$$p_w = \gamma_w \cdot H_{wtr}$$

Result: Backwater $p_{w,back} = 4.91 \text{ kPa}$, Frontwater $p_{w,front} = 1.47 \text{ kPa}$

D. Uplift Water Pressure under Base:

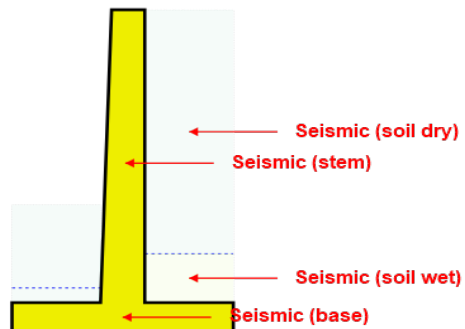


• Base Uplift Pressure Equation:

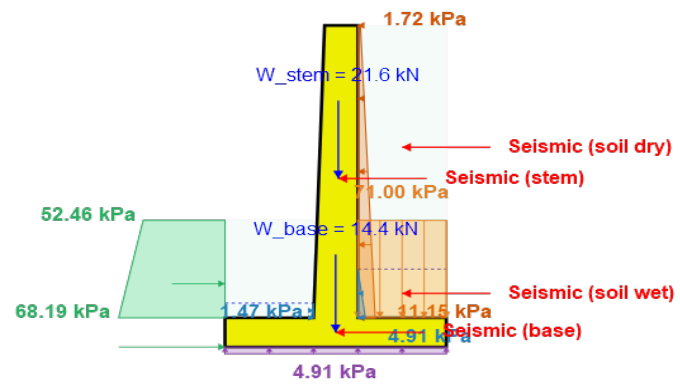
$$p_{uplift} = \gamma_w \cdot H_{wtr}$$

Result: $p_{uplift} = 4.91 \text{ kPa}$

E. Pseudo-Static Seismic Forces:



F. Total / Combined Load System:



4. Stability Verification

Safety factor analysis for overall retaining wall sliding, overturning, and slope failures:

Stability Verification Equations:

- Sliding safety factor equation:

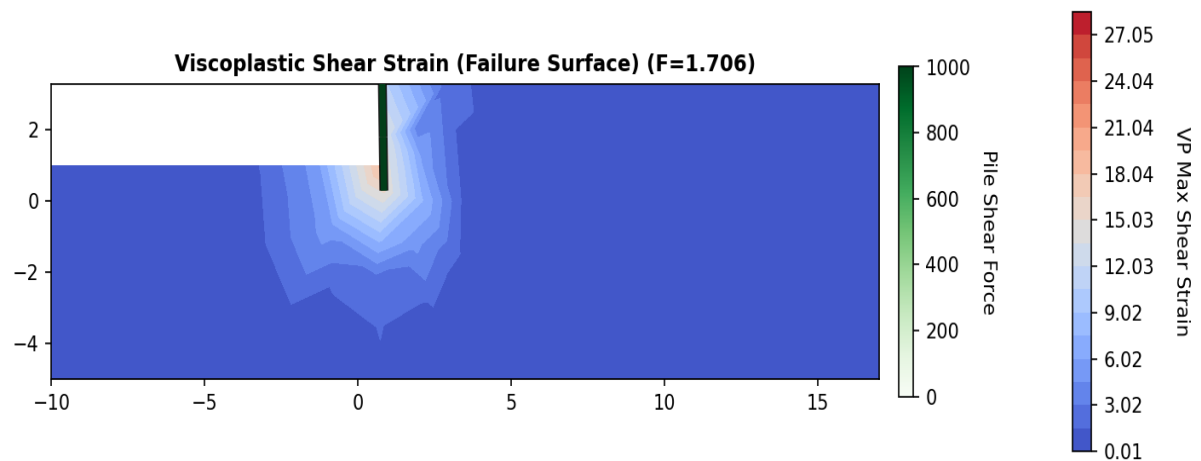
$$FS_{sliding} = \frac{\sum R_{resisting}}{\sum F_{driving}} = \frac{R_{sliding} + P_{passive}}{F_{driving}}$$

- Overturning safety factor equation:

$$FS_{overturning} = \frac{\sum M_{resisting}}{\sum M_{overturning}} = \frac{M_{res}}{M_{ot}}$$

Sliding Safety Factor Check	FS = 72.495	Req >= 1.500	PASS
Overturning Safety Factor Check	FS = 4.297	Req >= 1.500	PASS
Slope Stability SSRM Safety Factor Check	FS = 1.706	Req >= 1.1	PASS

2D SSRM Slip Surface Contour:



5. Bearing Capacity & Consolidation Settlement

Footing geotechnical bearing capacity design check and consolidation settlement breakdown:

Bearing Stress & Settlement Equations:

- Bearing contact stress:

$$\sigma_{max,min} = \frac{R_v}{B} \left(1 \pm \frac{6e}{B} \right)$$

- Bearing safety factor:

$$FS_{bearing} = \frac{q_{ult}}{\sigma_{max}}$$

- Consolidation settlement:

$$S_c = \frac{C_c \cdot H_c}{1 + e_0} \log \left(\frac{\sigma'_{vo} + \Delta\sigma}{\sigma'_{vo}} \right)$$

Ultimate soil bearing capacity (q_ult)	4506.35 kPa
Max bearing stress (sigma_max)	52.98 kPa
Bearing capacity Safety Factor	FS = 85.05 (Req >= 3.0)
Consolidation Settlement	94.00 mm (Allowable: 153.33 mm) -> PASS

6. Finite Element Settlement & Base Displacements

Toe Settlement: **23.43 mm**

Heel Settlement: **18.14 mm**

Differential Settlement: **41.57 mm** (Allowable: 50.00 mm) -> **PASS**

Rotational Tilt: **1.1907°**

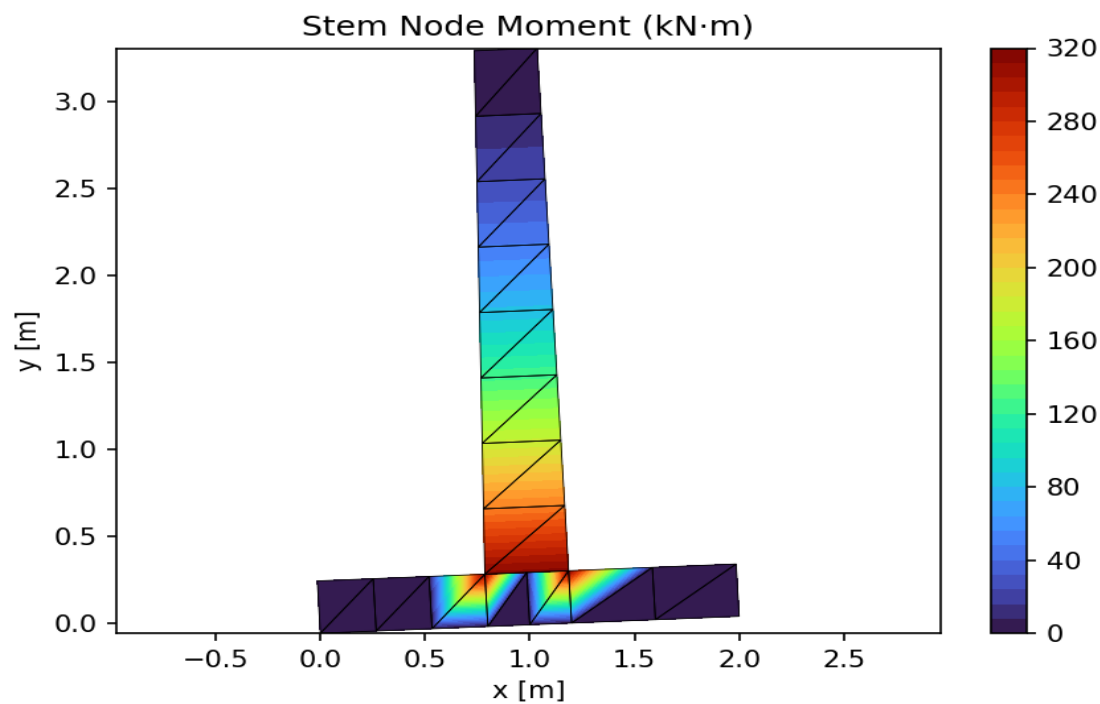
Footing Base Nodes Settlement Data:

Node	X Position (m)	Disp X (mm)	Disp Y (mm)	Settlement (mm)
1	0.000	0.0000	-23.4316	23.4316
2	0.267	-0.0076	-17.8998	17.8998
3	0.533	-0.0112	-12.3744	12.3744
4	0.800	-0.0030	-6.8338	6.8338
5	1.000	-0.0021	-2.6399	2.6399
6	1.200	-0.0142	1.5512	1.5512
7	1.600	-0.0277	9.8490	9.8490
8	2.000	-0.0316	18.1381	18.1381

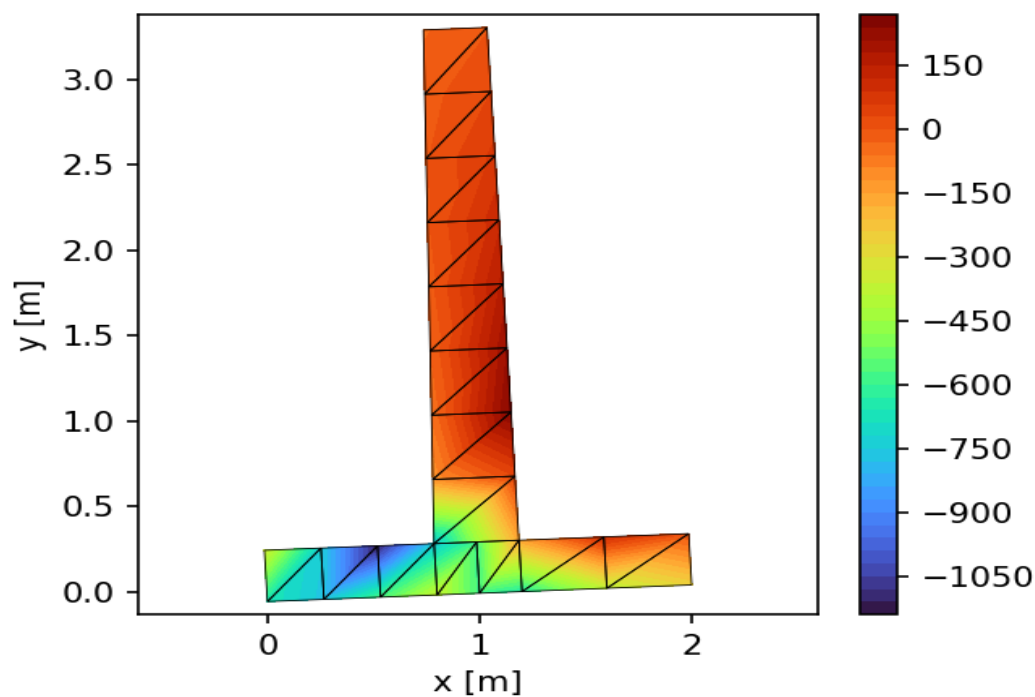
7. Finite Element Stress & Bending Moment Contours

2D internal bending moments and finite element stress contours plotted on the retaining wall concrete cross-section:

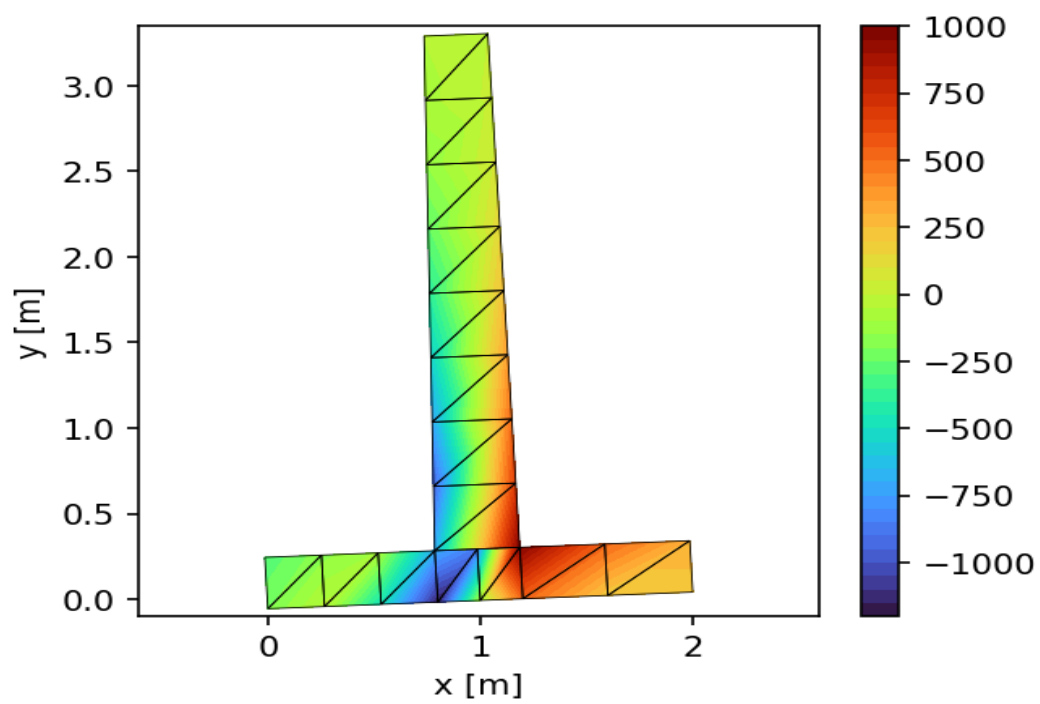
Stem Node Bending Moment Diagram (kN·m):



FE Stress Contour σ_{xx} (kN/m²):



FE Stress Contour σ_{yy} (kN/m²):



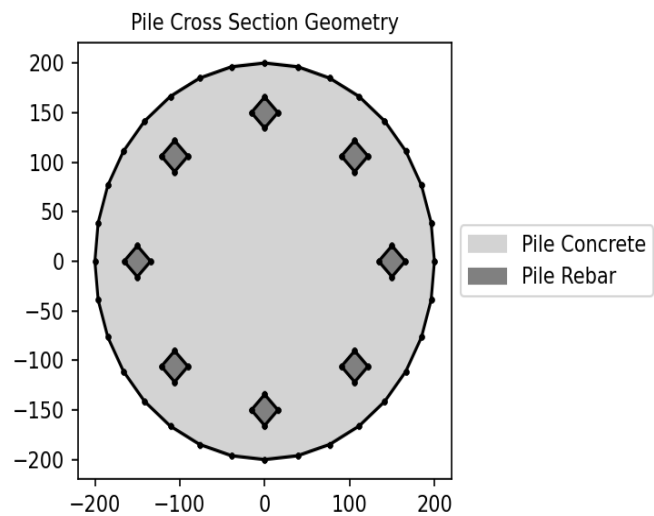
8. Pile Foundation Design & Verification

Geotechnical capacity, design parameters, and structural envelopes of the foundation piles:

Pile Cross Section Geometry & Material Details:

Parameter Name	Symbol	Value
Pile Material Type	-	Concrete
Pile Cross-section Shape	-	Circle
Pile Diameter	d_pile	0.400 m
Concrete Cover	d_c	50.0 mm
Main Reinforcement	-	8 D25

Pile Concrete Section Drawing:



Pile Structural Capacity Envelope limits:

Capacity Metric	Allowable Value
Axial Compression Capacity	215.47 kN
Axial Tension Capacity	113.14 kN
Lateral Shear Capacity	4320.00 kN

Demands Verification checklist:

Toe Pile Axial demand: **36.84 kN** -> **PASS**

Toe Pile Lateral shear demand: **48.10 kN** -> **PASS**

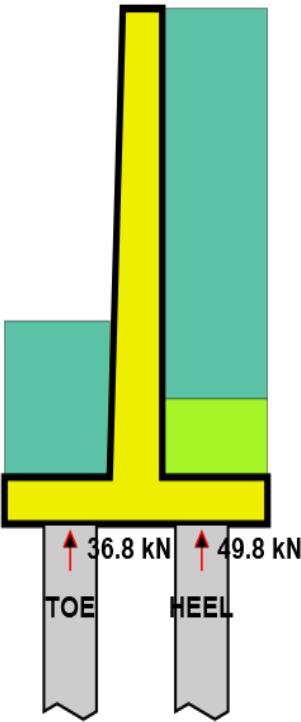
Toe Pile Interaction Envelope check: **PASS**

Heel Pile Axial demand: **49.75 kN** -> **PASS**

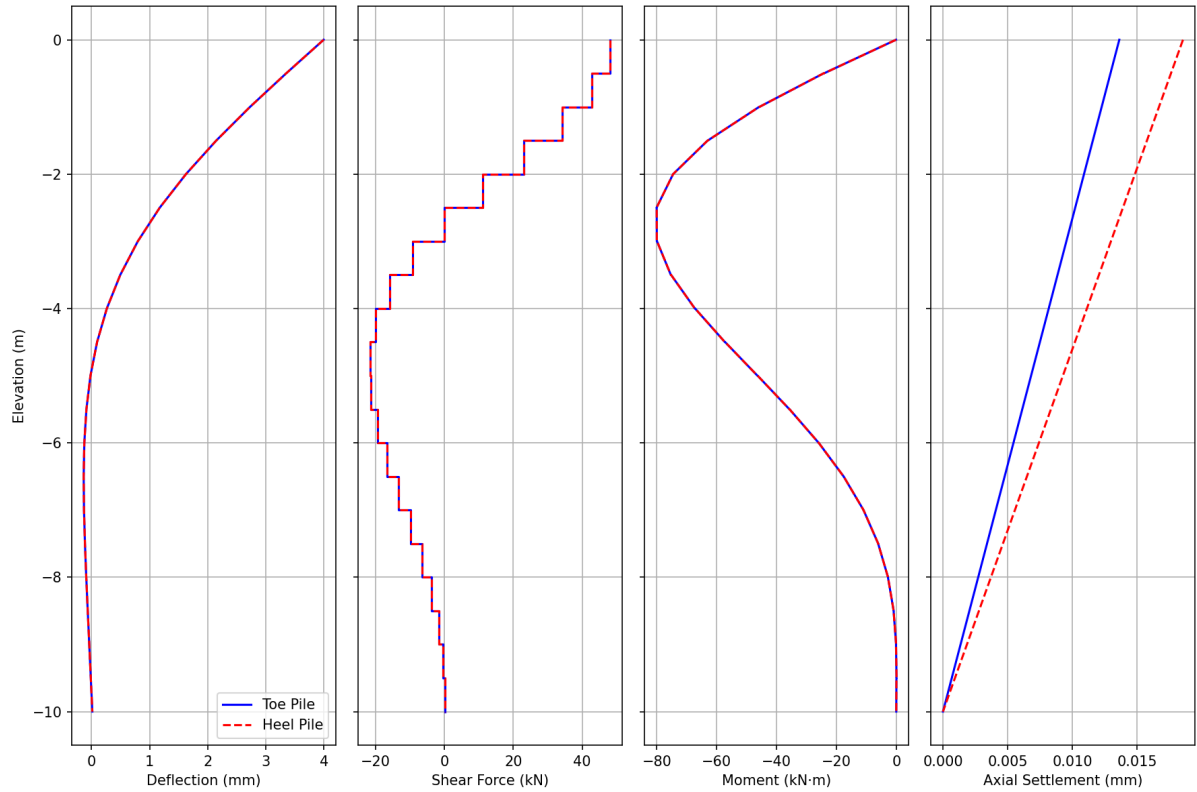
Heel Pile Lateral shear demand: **48.10 kN** -> **PASS**

Heel Pile Interaction Envelope check: **PASS**

Pile Foundation Schematic under Base:



Pile Winkler Analysis Elastic Profiles:



Moment-Axial Interaction Envelope:

